

RAJEEV SHARMA

SOFTWARE ENGINEER | PYTHON · JAVA · BACKEND · AI/ML

Berlin, Germany | +49 176 6120 4130 | rjsharma.ing@gmail.com
[linkedin.com/in/rjsharma](https://www.linkedin.com/in/rjsharma) | github.com/Rjsharma95 | rajeev-dev.vercel.app

PROFILE

Computational Engineering graduate with nearly two years of R&D experience across Java, Python, Linux-based systems, sensor interfaces and applied machine learning. At SectorCon, worked on Java communication and data-conversion tasks, JVM testing/debugging and system integration, and later completed a bachelor thesis that independently covered the full ML workflow from real-world pulse-wave acquisition to CNN training and Akida simulation. Currently strengthening backend and full-stack engineering through hands-on projects in Flask, FastAPI, React and planned reimplementations in Spring Boot and Django.

TECHNICAL SKILLS

Programming

Backend & Web

Data & AI/ML

Signal Processing

Systems & Embedded

Tools & Platforms

Python, Java, SQL, JavaScript, C, Bash, MATLAB, R

Flask, FastAPI, REST APIs, SQLite, React, Tailwind CSS, Vite; Spring Boot and Django in learning roadmap

NumPy, pandas, SciPy, scikit-learn, TensorFlow/Keras, CNNs, data preprocessing, feature engineering, model evaluation, class-imbalance handling, R/Shiny

FIR filtering, segmentation, normalization, time/frequency-domain analysis, CWT, STFT

Linux, Raspberry Pi, SPI, UART, JVM remote execution/debugging, byte-level data handling, JSON conversion

Git/GitHub/GitLab, JUnit, pytest, Docker basics, LabVIEW, CVAT, Anaconda, AWS EC2 basics, Azure basics; ROS/ROS 2, Kafka, MQTT and OPC UA fundamentals

PROFESSIONAL EXPERIENCE

Working Student - Research & Development | SectorCon Ingenieurgesellschaft mbH, Berlin

11/2023 - 09/2025

- Developed Java software in a Linux-based R&D environment involving hardware interfaces, serial communication and application-level data processing.
- Researched and worked with SPI and UART concepts; implemented Java functionality to retrieve and evaluate modem and Wi-Fi status using Linux system commands.
- Adapted an existing legacy implementation to build publish/subscribe functionality in Java and collaborated on SPI data transmission, reconstructing byte-level data and converting it to structured JSON.
- Created JUnit tests and researched/configured remote JVM execution and debugging for Java applications running on remote Linux systems.
- Used Git/GitLab and prepared technical documentation, reports and engineering presentations while working independently within an interdisciplinary team.
- Completed the bachelor thesis below at SectorCon in cooperation with TU Berlin as part of the R&D role.

BACHELOR THESIS | GRADE 1.3

Signal Quality Analysis of Medical Sensor Signals Using Neural Networks and Implementation on a Neuromorphic Chip

TU Berlin in cooperation with SectorCon | 2025

- Independently planned and built an end-to-end ML pipeline from real-world pulse-wave acquisition and preprocessing to CNN training, evaluation and preparation for neuromorphic deployment.
- Collected 250 Hz pulse-wave data with a piezoelectric pressure sensor; implemented filtering, feature analysis, labeling, segmentation, Z-score normalization and 2D time-frequency representations using wavelet transforms.

- Designed a TensorFlow/Keras Conv2D classifier adapted to Akida constraints, addressed class imbalance with random oversampling and evaluated performance using accuracy, precision, recall, F1, ROC/AUC and confusion matrices.
- Achieved 81.49% test accuracy; quantized and converted the model with QuantizeML/cnn2snn and evaluated it in the Akida Runtime simulation environment at 79.53% accuracy. No physical chip deployment was performed.

SELECTED ACADEMIC PROJECTS

Engineering Data Analytics - Automotive Failure Analysis | R, Shiny, R Markdown, Plotly

TU Berlin | Team of 4

- Contributed across data wrangling, integration, statistical analysis and development of an interactive Shiny web application for an automotive component-failure case study using production, registration and geographic data.

Autonomous Mobile Cleaning Robot - Concept Design

TU Berlin | Team of 2 | 2025

- Designed a modular autonomous wet-cleaning robot concept covering requirements engineering, safety, differential-drive motion, cleaning subsystems, Raspberry Pi/LiDAR/camera-based navigation, ROS-oriented integration, CAD layouts and route planning for a 534 m² test hall.

AI Assistance for In-Process Manufacturing Quality Control

TU Berlin | Team of 4

- Annotated manufacturing-related image/video data in CVAT for a planned AI-assisted quality-control system using process data, video and thermographic imagery; the broader project was not completed.

PERSONAL SOFTWARE PROJECTS

Job Application Tracker API | Python, Flask, SQLite | In progress

- Building a CRUD backend for job-application tracking with persistent storage, status validation and tests; planned reimplementations in Spring Boot and Django to deepen backend framework knowledge.

wohnmatch-AI | Python, FastAPI, React | In progress

- Developing a flagship housing-search application with a FastAPI backend; planned capabilities include listing ranking, preference-based matching, application workflow automation and tracking.

Personal Portfolio | React, JavaScript, Tailwind CSS, Vite

Deployed: rajeev-dev.vercel.app

- Built and deployed a responsive portfolio website with reusable React components and mobile-friendly layouts.

EDUCATION & ONGOING LEARNING

B.Sc. Computational Engineering | Technische Universität Berlin

2017 - 2025 | Final grade 2.8

Focus: programming, computer engineering, AI/ML, data analysis and automation engineering.

Studienkolleg, T-Kurs | Technische Universität Berlin

2015 - 2016

CS50x: Introduction to Computer Science | Harvard University

Online course | In progress

LANGUAGES

German C1 | English C1 | Nepali native | Hindi fluent | French A1